# IPO Portfolio Optimization with GRU-Based Allocation

STAT 4830 — Spring 2026

Shivani Shreedhar, Olivia Liu, Oceana Zhu

# Problem Statement

## Portfolio Allocation

We optimize daily portfolio weights between two asset classes: a market index (82% S&P 500, 18% Dow Jones, proxied via SPY and DIA from CRSP) and a custom IPO index — a market-cap-weighted basket of recent IPOs sourced from SDC Platinum, where each stock is held for 180 trading days post-IPO.

## Why This Matters

IPOs exhibit 3–4× higher volatility than the broad market, with mean reversion in the first six months and information asymmetry between institutional and retail participants. Static allocations ignore regime shifts (COVID crash 2020, IPO boom 2021, correction 2022).

**Constraints:** Long-only, fully invested, no shorting or leverage, daily rebalancing.

# Technical Approach: Loss Function

The model minimizes a composite differentiable loss with six modular, tunable components:

$$\mathrm{L} = -\mu_p + \lambda_{CVaR} \cdot CVaR + \lambda_{Var} \cdot Var(r_p) + \lambda_{excess} \cdot \max(0,\, \sigma_{ann} - \sigma_{target}) + \lambda_{turnover} \cdot \|w_t - w_{t-1}\|_1 + \lambda_{path} \cdot \|w_t - w_{t-1}\|_2^2$$

## Return Maximization

Negative mean return objective

## Tail Risk Control

CVaR at 5% level via soft-sorting approximation

## Variance Penalty

Volatility excess penalty (target 25% annual vol)

## Turnover Penalty

L1 norm and path stability (L2 squared)

# Technical Approach: Model Architecture

### Input Layer

84-day rolling window of 4 features: market return, IPO return, 21-day rolling volatility, VIX level

2

### GRU Network

Gated Recurrent Unit: input=4, hidden=64, 1 layer

3

### MLP Head

2-layer MLP (64→64→2)

4

### Softmax Output

Portfolio weights $w = [w_{market}, w_{IPO}]$ with $w_i \geq 0$, $\sum w_i = 1$

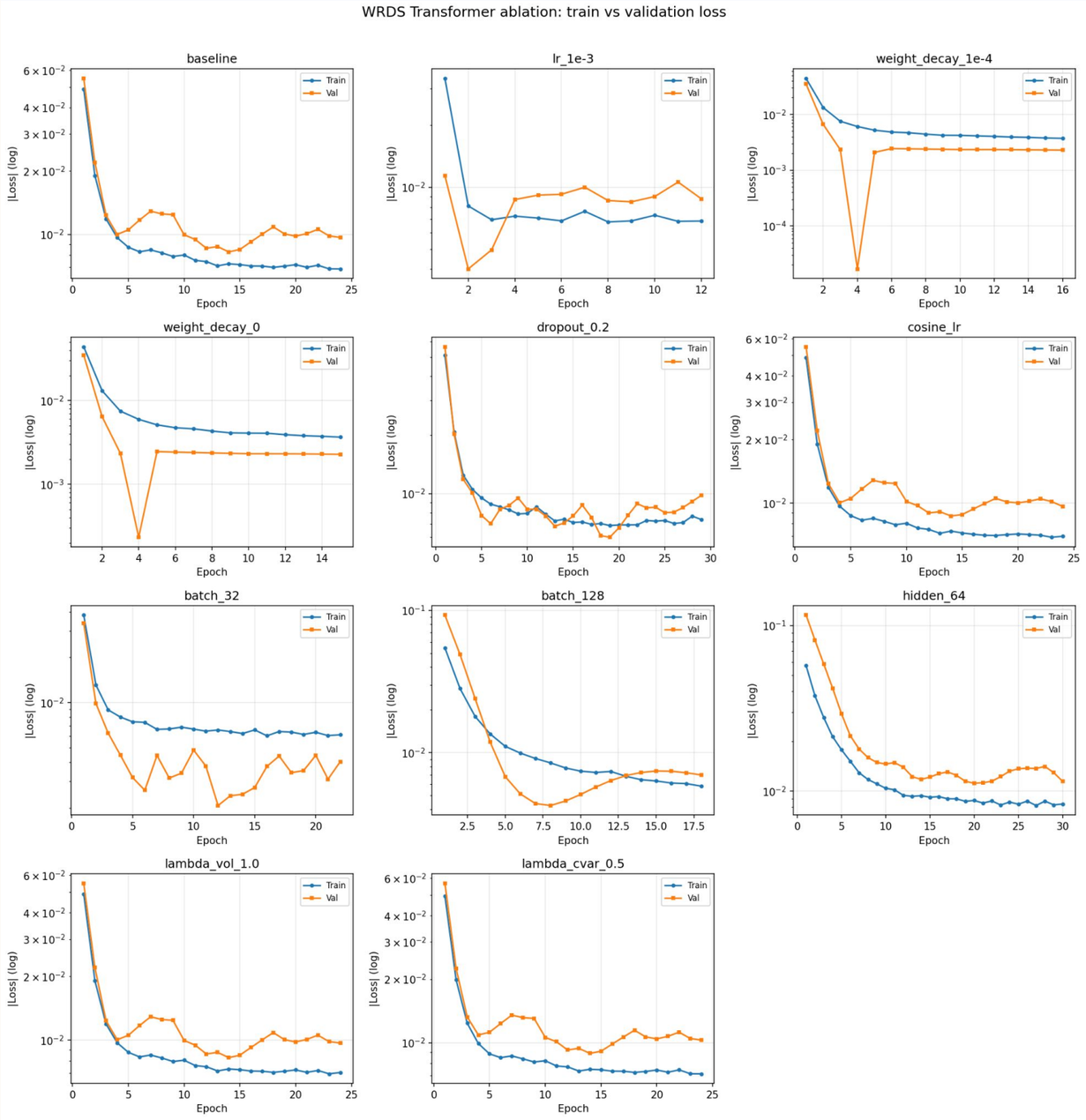
The GRU extracts temporal patterns from the rolling window; the softmax output guarantees valid simplex-constrained weights. GRU was chosen over LSTM for fewer parameters and faster training on our small dataset (~1,100 samples).

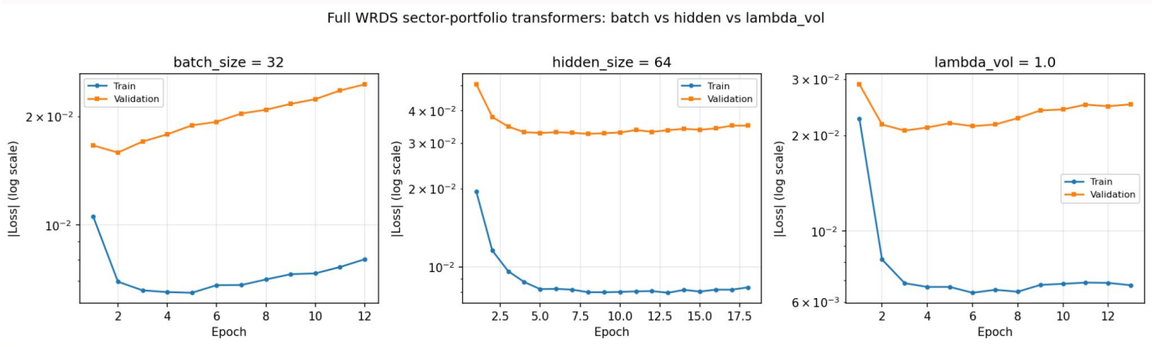
# Hyperparameter Tuning Results

288-configuration grid search over all six loss penalty weights, target volatility, and window length. ~19 minutes total runtime.

| Parameter | Previous (Hardcoded) | Tuned (Best) |
|---|---|---|
| λ_turnover | 0.01 | **0.0025** |
| λ_path | 0.01 | **0.01** |
| λ_vol | 0.5 | **0.25** |
| λ_cvar | 0.5–1.0 | **0.25** |
| window_len | 126 days | **84 days** |

# Graphs (Transformer)



# Hyperparameter Tuning Results (Transformer)

| Parameter | Previous (Hardcoded) | Tuned (Best) |
|---|---|---|
| Batch Size | 32 | **32** |
| Hidden Size | 64 | **64** |
| λ_vol | 0.5 | **1.0** |
| Learning Rate | 1e-3 | **3e-4** |
| Weight Decay | 1e-5 | **1e-2** |

# Returns (Transformer)

`batch_32` (batch_size = 32)

| Sector | AnnRet | Sharpe | MaxDD |
|---|---|---|---|
| Unknown | 34.61% | 1.51 | -22.27% |
| Health_Care | 35.98% | 1.63 | -17.38% |
| Information_Technology | 26.92% | 1.01 | -32.09% |
| Financials | 10.51% | 0.60 | -25.22% |
| Consumer_Discretionary | 12.73% | 0.56 | -42.63% |
| Industrials | 35.81% | 1.40 | -22.40% |
| Energy | 8.75% | 0.55 | -26.28% |
| Communication_Services | -10.21% | -0.21 | -57.89% |
| Real_Estate | 5.79% | 0.33 | -45.16% |
| Consumer_Staples | 5.48% | 0.35 | -30.33% |
| Materials | -8.62% | -0.10 | -47.61% |

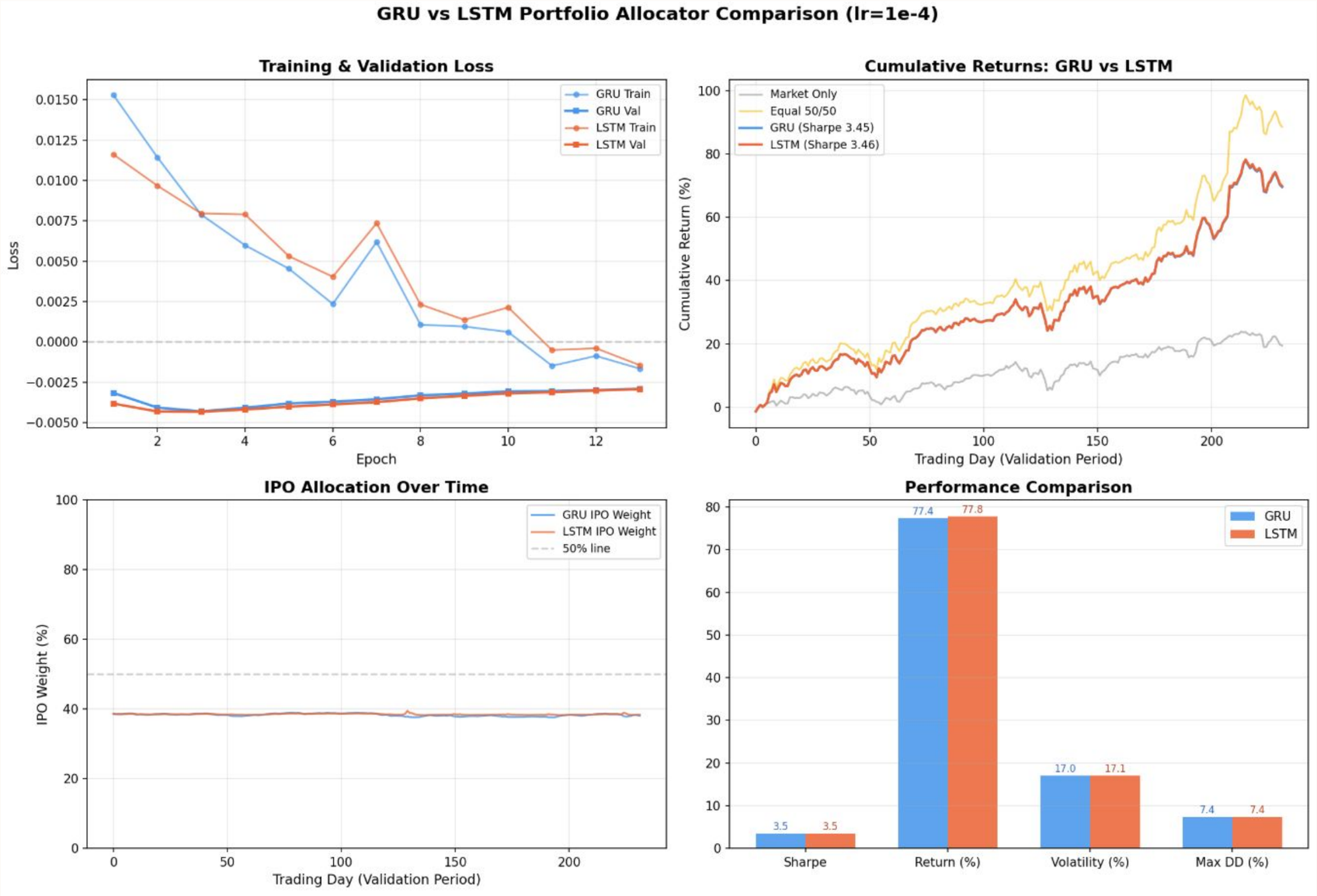
`lambda_vol_1.0` (lambda_vol = 1.0)

| Sector | AnnRet | Sharpe | MaxDD |
|---|---|---|---|
| Unknown | 39.81% | 1.57 | -22.63% |
| Health_Care | 39.92% | 1.71 | -17.07% |
| Information_Technology | 29.81% | 1.01 | -36.96% |
| Financials | 10.50% | 0.60 | -25.31% |
| Consumer_Discretionary | 12.57% | 0.53 | -47.23% |
| **Industrials** | **-70.52%** | **-7.26** | **-97.09%** |
| Energy | 8.39% | 0.53 | -27.43% |
| Communication_Services | -18.08% | -0.34 | -71.31% |
| Real_Estate | 7.00% | 0.37 | -40.52% |
| Consumer_Staples | 3.72% | 0.27 | -34.56% |
| Materials | -7.45% | -0.08 | -46.01% |

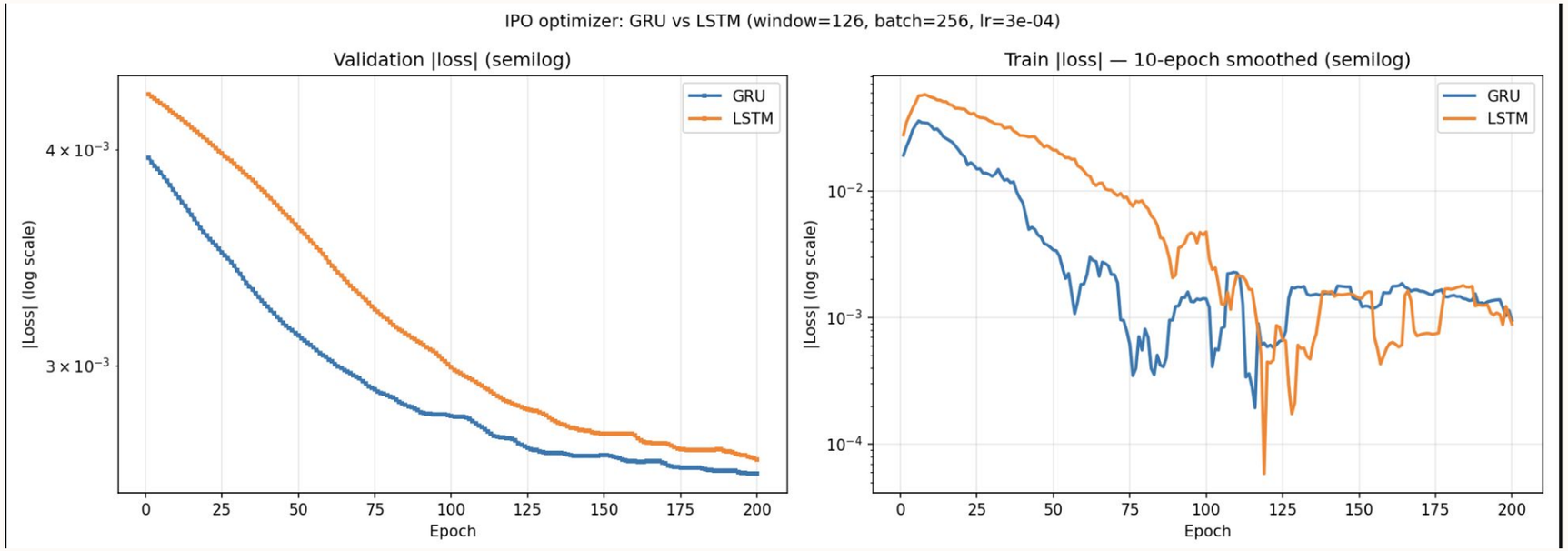
`hidden_64` (hidden_size = 64)

| Sector | AnnRet | Sharpe | MaxDD |
|---|---|---|---|
| Unknown | 45.19% | 1.61 | -22.99% |
| Health_Care | 49.28% | 1.85 | -16.46% |
| Information_Technology | 30.37% | 1.01 | -37.91% |
| Financials | 10.26% | 0.56 | -26.83% |
| Consumer_Discretionary | 12.05% | 0.49 | -53.40% |
| Industrials | 45.28% | 1.45 | -23.44% |
| Energy | 7.34% | 0.45 | -30.71% |
| Communication_Services | -21.30% | -0.38 | -75.62% |
| Real_Estate | 2.41% | 0.27 | -54.95% |
| Consumer_Staples | 2.35% | 0.22 | -37.83% |
| Materials | -14.35% | -0.18 | -54.82% |

# Graphs (GRU vs LSTM)



# Results (GRU vs LSTM)




| Model | Sharpe | Max drawdown | Total return |
|---|---|---|---|
| GRU | 3.47 | −7.19% | 78.35% |
| LSTM | 3.55 | −7.34% | 87.63% |

# Failure Mode Analysis

## Static Allocation Collapse (Root Cause)

λ_path=0.01 & λ_turnover=0.0025 penalized any weight change. The GRU minimized loss by outputting a near-constant 68/32 split — avg daily turnover 0.0031, identical weights every day. The model learned a static policy, not a dynamic one.

## Regularisation Dominates Return Signal

With batch_size=32, gradient noise was high. The path/turnover terms had smooth, easy-to-minimise gradients while the return signal was noisy. Optimizer converged to the trivial fixed-weight solution in <5 epochs.

## Validation Leakage via Hyperparameter Search

288-config grid search selected the config maximising validation Sharpe. The reported Sharpe (3.27 → 3.60) is optimistic. A true held-out test set is needed to measure generalisation honestly.

## Survivorship Bias in IPO Index

The IPO index includes only tickers with price data in CRSP. Failed / delisted IPOs within the holding window are systematically excluded, inflating the IPO index return (192% in 2024).

## Validation Period Regime Risk

Jan–Dec 2024 contained five distinct regime shifts (Apr sell-off, Aug VIX-65 carry unwind, Sep Fed cut, Nov election rally, Dec hawkish pivot). A static model cannot adapt; a dynamic model must learn these signals from 84-day windows.

## Remaining Gap vs. Equal Weight (Post-Fix)

After fixing λ_path=0.0001 & λ_turnover=0.0, Sharpe improved to 3.60 but still trails equal-weight (3.65). The GRU has not yet learned to time the market. IPO weight rose to 46% but is still near-static; true dynamic allocation requires richer regime features.

# Observations

### Valid simplex-constrained weights

The model trains end-to-end and produces valid weights (69% market / 31% IPO average)

### Strong Sharpe ratio

Sharpe of 3.27 is strong, but validation set was also used for hyperparameter selection — no configs were evaluated on truly unseen data

### Four informative features

Real VIX replaced the constant 20.0 placeholder; the model now receives four informative features

### Shorting Stocks

Shorting stocks although slughlty increased returns, it had worse validation and training loss

### Context Window

Changing the context window did not have a big impact on our model

# Areas for Improvement + Next Steps

## Clear Sector Divisions

- Try the LSTM & GRU with sector divisions
- Sector “Unknown” is the largest sector, even after pulling industry classifications (GICS) from WRDS Compustat data

## Model underperforms naive 50/50

- Improvements include re-scaling or re-weighting loss terms, explicit benchmark-relative terms (e, transaction-cost and turnover models, and walk-forward or nested validation so “beating 50/50”

## Utilize GPU credits

- Each run of the model takes ~1 hour on the 2010-2024 data, as we consider increasing the dataset, this time will only increase
- Running the model on the GPU would allow us to iterate faster and over a larger dataset

## Addressing Loss Spikes

- Normalizing the size of the gradient
- Make step sizes non-constant